Literature Review: Optimal Sensor Placement for Wastewater Network Monitoring

Source classification supporting Project Objective PO1 and Deliverable D1

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Abstract

This review supports Project Objective PO1 and Deliverable D1. It examines nineteen sources spanning five bodies of work relevant to the project: the formulation of the sensor placement problem itself; the detection of blockages and overflows from sensor data, including current Australian utility practice; the network attributes that determine where monitoring is most warranted, covering both pipe condition and inflow and infiltration; the reliability, cost and information value of sensor hardware once installed; and the broader evidence base on AI-driven sewer condition assessment against which the project's parameter catalogue can be cross-checked. Each source is classified by optimisation formulation, objective function, input parameters, network type, and applicability to a gravity trunk sewer network. Six placement formulations are compared: deterministic global optimisation with decoupled hydraulics, topological ranking by in-relevance harmonic centrality, coverage optimisation from network topology and elevation by genetic algorithm, data-driven sparse sensing, information-theoretic placement by mutual information and by re-clustering scored with joint entropy, and goal-oriented Bayesian optimal experimental design. Three detection studies (a utility survey, a level-based discriminant and a paired acoustic method) show that what a sensor "covers" depends on the detection rule it runs, and six technology studies show that sensor reliability, cost and information content vary by device and by location. The reviewed placement formulations are reusable, but their objective functions only partly are: the one formulation built for blockage detection in gravity sanitary sewers, from an Adelaide research group, treats typical water depth as a fixed input, and none models how coverage changes as growth increases the connected load, which is the project's primary application. Eight gaps are recorded, of which the absence of an agreed observability measure for rising sewer level under increased connected load remains the principal open question with the client.

Keywords: wastewater network; sewer monitoring; optimal sensor placement; mutual information; genetic algorithm; sparse sensing; complex network theory; blockage detection; soft sensors; Bayesian networks; sensor reliability; SA Water.

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1 Scope and Purpose

This review supports Project Objective PO1 and Deliverable D1. It examines nineteen sources that together span five bodies of work relevant to the project: the formulation of the sensor placement problem itself; the detection of blockages and overflows from the data a placed sensor produces; the identification of the network attributes that determine where monitoring is most warranted; the practical reliability, cost and information value of the sensor hardware once installed; and the broader evidence base on AI-driven sewer condition assessment against which the project's parameter catalogue can be cross-checked. Each source is classified by optimisation formulation, objective function, input parameters, network type, and applicability to a gravity trunk sewer network, consistent with the classification scheme committed to in PO1. This revision extends the first issue, which covered six sources, to every research paper held in the author's section of the team library.

The sources fall into five groups.

- **Placement formulations** (Section 2). Simone et al. (2023) compare a deterministic optimisation formulation against a topological one for monitoring planning in sewer networks; Ninh et al. (2025) optimise the coverage of water level sensors for blockage detection in two real gravity sanitary sewers using network topology and elevation alone; Ding et al. (2026) formulate placement as a data-driven sparse sensing problem for system-wide flow reconstruction; Crowley et al. (2025) maximise the mutual information between sensor observations and the state of large sewer networks; Wang et al. (2023) place sensors at the centroids of clustered node behaviour and refine the layout by re-clustering on utility priorities; and Cheng et al. (2026) show, through Bayesian optimal experimental design, that the optimal location changes with the prediction the sensor is meant to inform. These inform the placement method (D3).
- **Blockage and overflow detection** (Section 3). Drenoyanis and Prackwieser (2022) survey the methods Sydney Water uses to detect wastewater blockages; Li et al. (2023) identify and position blockages from paired level measurements; and Ninh et al. (2026) locate and size blockages within a pipe using paired acoustic sensors. These define what "detection" means at a placed sensor and so constrain the placement objective (D3).
- **Network attributes** (Sections 4–6). Malek Mohammadi et al. (2020) review the factors influencing sewer pipe condition; Ma et al. (2025) quantify those factors with a data-driven Bayesian network; Ma et al. (2026) review 71 studies (2016–2025) on AI-driven sewer condition assessment; and Wang et al. (2024) and Zhang et al. (2020) use machine learning to identify inflow and infiltration from routine flow monitoring. These inform the placement parameter catalogue (D2).
- **Sensor technology and data value** (Section 7). Alshami et al. (2023) test flow and level sensor performance in small- and medium-diameter sewers in Hong Kong; Utepov et al. (2023) prototype a low-cost IoT chamber monitor and bench-test four distance sensors; Lin et al. (2025a, 2025b) derive flow from level measurements with soft sensors; and Wani et al. (2015) quantify the information carried by cheap binary sensors. These bear on sensor type selection and the cost and information trade-off analysis (D5), and on the reliability assumptions embedded in the placement objective itself.
- **Cross-checks and explainability.** Ma et al. (2026) also cross-checks the parameter catalogue (D2) against a much larger evidence base and surfaces explainability techniques relevant to the project's explainability requirement (R4).

2 Formulations of the Sensor Placement Problem

2.1 Deterministic optimisation

Simone et al. (2023) formulate sensor placement as a mixed-integer problem solved to global optimality using the SCIP global optimisation solver. Each node of the network carries a binary installation variable, a second binary variable records whether a contaminant is detected at a node under a given injection scenario, and a cardinality constraint fixes the total number of sensors. The objective maximises network reliability, defined as the ratio of detected node-scenario pairs to the total number of detectable pairs across all scenarios.

The methodological feature most relevant to this project is the decoupling of the optimisation from the hydraulic model. Flow velocities and pollutant concentrations are computed once in SWMM and enter the optimisation as fixed parameters, so the solver does not invoke a hydraulic simulation at each candidate configuration. This is the same architectural choice that makes the project's own proposed approach tractable within the project timeframe: simulation is used to characterise the network, not to search it. Reported solve times were under one second even for the larger of the two test networks.

2.2 Topological optimisation via complex network theory

The second formulation in Simone et al. (2023) abstracts the network as a directed graph and ranks nodes by in-relevance-based harmonic centrality, a variant of harmonic centrality that weights each node by its intrinsic relevance (here, lateral inflow) and that accumulates only information arriving at a node. The justification offered for harmonic centrality in this setting is that transport in a gravity sewer follows shortest paths determined by slope, so the metric's underlying assumption matches the physics of the system. No simulation is required; only topology and inflow data are needed.

Two results bear directly on the project. First, the ranking is normalised to a $[0, 100]$ range and the authors adopt values above 20 as the criterion for candidate sensor locations, which yields a small candidate set on a real network. Second, head nodes return a null metric value, because information does not arrive at them, and are therefore excluded as candidates. This second property is a material limitation for the project's target application and is discussed in Section 8.2.

2.3 Coverage optimisation from network topology and elevation

Ninh et al. (2025) is the formulation closest to the project's setting. It was developed at the University of Adelaide, motivated by the smart sewer monitoring already deployed in urban South Australia, where ultrasonic water level sensors have been installed in sanitary sewers to monitor blockages, and its stated aim is to find the minimum number and best locations of level sensors for proactive detection of dry-weather overflow. Like the topological formulation above, it needs no hydraulic model; unlike it, the coverage of a sensor is tied to a physical event rather than to a centrality score.

The formulation starts from an observation in the utility's own data. A sensor upstream of a blockage records a clear rise in level, but a sensor downstream of it often shows no clear signal, because maintenance holes are junctions of several branches, household discharge is intermittent and can fall to zero off-peak, and readings are low-frequency or not synchronised. The method therefore places sensors only to detect blockages in the pipes downstream of them. A sensor at maintenance hole s is said to cover a downstream node if a blockage just below that node would raise the level at s before the node itself overflows:

$$Z_0 + H_d + \epsilon > Z_s + H_s,$$

where Z_0 is the invert elevation of the downstream node, H_d the depth threshold at that node that would trigger an overflow alert, Z_s the invert elevation at the sensor, H_s the typical (no sudden discharge) depth at the sensor, and ϵ an elevation error term, set to zero in the study. H_d was set at 70% of the node depth from the level rises observed during blockage events, and the authors note that a lower H_d means earlier warning but more sensors. A breadth-first search lists the "downstream-dependent" nodes each maintenance hole covers; only maintenance holes (not inspection holes) are candidates, on work health and safety grounds.

A genetic algorithm (population 1,000, 200 generations) then solves two problems: the minimum number of sensors that covers every coverable node, and the locations of a fixed number of sensors that maximise the total pipe length covered, which models a budget released in investment phases. A multistage variant fixes the sensors of earlier phases and optimises only the new ones. Two real smart sanitary sewer systems scheduled for extension were studied; the paper does not name their location and describes the data as confidential. Both are gravity sewers of 150 mm vitreous clay and uPVC pipe: System I is flat (average slope about 1.4%; 103 maintenance holes, 266 pipes, 12.5 km, 10 sensors installed) and System II is steep (189 maintenance holes, 322 pipes, 11.3 km, 82 ultrasonic sensors installed).

Three results matter for the project. First, terrain dominates the answer: full coverage needed 64 sensors in the flat system but 170 in the steep one, where 142 of 189 maintenance holes covered no downstream node at all, and the authors conclude that level-rise detection is a weak basis for coverage in steep systems. Second, ten optimised sensors in System I covered 2,954 m of pipe against 2,069 m for the ten sensors the utility had placed from blockage history, about 43% more; the utility's layout looked well spaced but five of its sensors observed only one or two short pipes. Third, extending the existing ten sensors with ten optimised ones covered 4,555 m, whereas optimising all twenty together covered 5,062 m, so sensors placed early without a network-wide view carry a lasting cost unless they are relocated. The authors also find that optimal layouts are not unique, and they list as limitations that the objective is pipe length alone, with blockage history, environmental consequence, access and safety left for future work.

2.4 Data-driven sparse sensing via reduced-order reconstruction

Ding et al. (2026) formulate sensor placement as a data-driven sparse sensing (DSS) problem, coupling EPA-SWMM simulation with singular value decomposition (SVD) and QR factorisation with column pivoting to select the smallest set of nodes from which system-wide flow can be reconstructed. An SVD of a simulated training ensemble (250 scenarios spanning ten rainfall events and five imperviousness and five infiltration settings) yields a reduced temporal basis; QR factorisation with column pivoting on that basis then identifies the nodes whose measurements most efficiently reconstruct the full network state. As in Simone et al. (2023), the hydraulic model is used once, offline, to characterise the network rather than being re-invoked at each candidate configuration, the same architectural choice that keeps the approach tractable and that the project's own proposed method also relies on.

Applied to a 77-node storm sewer network in Duluth, Minnesota, three DSS-selected sensors reconstructed system-level flow with a mean Nash–Sutcliffe efficiency (NSE) of 0.949 across five observed storm events, rising to 0.993 at ten sensors; reconstruction of a 200-year design event improved from an NSE of 0.842 at one sensor to 0.994 at ten. Benchmarked against an exhaustive search (feasible for one to four sensors) and against 20,000-sample Monte Carlo random placements (five to ten sensors), the DSS-selected configurations tracked the global optimum closely (mean NSE gap of 0.002–0.013) and substantially outperformed random placement, whose reconstructions were frequently and severely negative in NSE. The framework degraded gracefully under multiplicative Gaussian measurement noise of up to 15% and, at moderate to high sensor counts, under single-sensor dropout, though dropout sensitivity was concentrated in a small number of structurally critical sensors whose importance was better explained by their redundancy, post-dropout conditioning and modal loading than by their QR selection

rank alone.

Two aspects of this formulation are directly relevant to the project's method (D3) and to Deliverable D5. First, DSS supplies an evaluation measure, reconstruction NSE against a held-out event, that is a genuine measure of network observability rather than a topological proxy, which is the gap Section 8.4 identifies as unresolved in most of the other placement formulations reviewed here. Second, the robustness testing built into the DSS evaluation, specifically its treatment of measurement noise and of sensor dropout as first-class experimental conditions, evaluates a placement's fault tolerance rather than assuming reliable readings at every instrumented node once installed; this partially closes, without fully resolving, the reliability gap raised in Section 7. The authors are explicit, however, that DSS is a linear reduced-order framework and that this limits its ability to represent the threshold-driven behaviour of overflow-prone nodes, which sit at or near zero flow under ordinary conditions and activate abruptly once a hydraulic threshold is exceeded; this is precisely the class of node the project's target application is designed to monitor, and the authors identify it as the framework's most significant limitation for exactly this reason.

2.5 Information-theoretic placement

Two sources score a layout by how much information its sensors collect about the network, rather than by detection of a specific event.

Mutual information for large networks. Crowley et al. (2025) pose placement as the maximisation of the mutual information between the states of all nodes and the noisy observations at the sensed nodes. Node states (flow or depth) are modelled as jointly Gaussian with a covariance estimated from simulated time series, sensor readings carry independent additive Gaussian noise, and the mutual information then reduces to a closed-form log-determinant of the observed covariance. Because the objective is submodular, a greedy search is guaranteed to reach at least 63% of the optimum; the authors' one-step modified greedy algorithm runs the greedy search from every possible first sensor and keeps the best result, and it can hold existing sensors fixed so that an established network can be extended. Two networks were studied: a 479-node foul network in the south of England serving about 12,000 people (two days of simulated dry-weather flow) and the 1,020-node combined network of Bellinge, Denmark (two years of SWMM simulation with observed rainfall). Placements were validated by how well a linear model and a general regression neural network estimated flow at every unmonitored node, measured by normalised mean square error (NMSE). In Bellinge, the placement gained at least 40% more mutual information than every alternative for 5 to 50 sensors and gave the lowest NMSE with both estimators up to 100 sensors, outperforming a rule-based layout that follows UK sewer-survey guidance (outfalls, junctions of large sub-catchments with the trunk sewer, and major trunk junctions); from 125 sensors upward the rule-based layout did better with the neural estimator. Optimal sensors tended to cluster in locally informative parts of the network, and placements for depth differed from those for flow. Run times were 18 minutes for 125 sensors on 479 nodes and about 14 hours for 250 sensors on 1,020 nodes.

Re-clustering for routine monitoring. Wang et al. (2023) reach a related answer by a simpler route. Node attributes generated by SWMM (maximum and average level, a chemical oxygen demand tracer, elevation and coordinates) are grouped by K-means, and a sensor is placed at each cluster centroid. Clusters are then split further according to a utility priority, either average closeness centrality (more sensors where the network is topologically central) or average node inflow (more sensors where more dry-weather flow, and so more population, drains). Layouts are scored by joint entropy (information collected, higher is better) and total correlation (redundancy between sensors, lower is better). On the 31-node combined sewer of the SWMM Applications Manual Example 8, three sensors gave the best balance and four or five sensors

were no better, because the extra sensors largely repeated what the first three already saw; the clustering layout also spread sensors across upstream, interceptor and downstream areas where two earlier methods had bunched them downstream. On a 786-node separate sewer serving an industrial park in Wenzhou, China, re-clustering generally beat plain clustering for the same number of sensors, and the authors recommend an initial 20 clusters. They concede that both re-clustering rules are simple and sometimes reduce information.

2.6 Objective-first design: Bayesian optimal experimental design

Cheng et al. (2026) do not study sewers, but they make an argument every other formulation here leaves implicit: a sensor location is optimal only relative to the prediction or decision it is meant to inform. Using Bayesian optimal experimental design, each of 2,576 candidate locations in an urban flood model of a Detroit corridor is assigned an expected information gain, either about the model parameters or about a stated flood prediction (point depths, regional averages or regional maxima at several lead times); a neural-network surrogate replaces the physical model to make the computation feasible. Different targets produced different spatial rankings: point targets favoured nearby locations while regional targets could favour distant ones, and learning the parameters did not translate evenly into better predictions. The expected information gain is the same mutual-information quantity Crowley et al. (2025) maximise, which places the two formulations on a common footing. The paper also separates information value from deployability: access, safety, power, communications, permission and maintenance restrict where a sensor can go, and the recommended workflow is to define the prediction target, screen the admissible set with field-verified restrictions, compute the information map, and only then rank locations.

2.7 Comparative findings

The two formulations in Simone et al. (2023) were applied to a benchmark network of 77 nodes, 79 links and one outfall, and to a real network of 531 nodes, 530 pipes and one outfall. On both, the node sets selected by the solver fell within the range of nodes ranked highest by the topological metric, or were adjacent to them, and lay along the same principal flow paths. The outfall was the highest-ranked node in both cases, as expected.

The two approaches diverge in one respect that matters for sensor selection. The optimisation solution is sensitive to the assumed sensor detection threshold: relaxing the threshold from 0.003 to 0.01 mg/L increased the number of sensors required for full coverage from four to eleven on the real network. The topological ranking, depending only on structure and inflow, does not change with sensor quality at all. For a project that must compare float switches against level sensors under an agreed observability measure (PO5), this is a direct finding: a purely topological criterion cannot by itself discriminate between sensor modalities, and some threshold- or capability-dependent component is required alongside it.

The authors conclude by proposing a two-stage procedure in which the topological metric pre-screens candidate nodes and the deterministic optimisation searches within that reduced set, on the grounds that deterministic solvers become computationally demanding at scale. They also note that the topological approach demands competent interpretation, since its results must still be checked for hydraulic plausibility.

A third point of comparison follows from placing Ding et al. (2026) alongside the two formulations above. Where the deterministic and topological methods in Simone et al. (2023) both target contaminant or intrusion detection and evaluate placement quality by a reliability or centrality score, DSS targets flow reconstruction and evaluates placement quality directly against held-out flow observations. This makes DSS's NSE-based evaluation the closest of the three to a genuine observability measure, but it inherits a different limitation: as a linear method, it is best suited to nodes whose flow varies continuously, and it is weakest exactly at the threshold-crossing

overflow nodes the project's overcapacity application must monitor.

Ninh et al. (2025) supplies the threshold-dependent component that the topological ranking lacks without giving up its independence from a hydraulic model. Its coverage condition depends on an alarm threshold (H_d) and on a typical depth (H_s), so the number and position of sensors change with the detection capability assumed, exactly as the deterministic formulation's did with its contaminant threshold. It is also the only formulation reviewed whose objective is blockage detection before overflow in a gravity sanitary sewer, and the only one tested against a utility's existing, experience-based layout, which it outperformed by about 43% in covered pipe length. Its weakness is the mirror image of DSS: it represents the threshold-crossing event well but assumes that the typical depth at each maintenance hole is known and fixed.

The information-theoretic formulations add two findings that the others do not test. First, more sensors are not proportionally more information: Wang et al. (2023) found four or five sensors no better than three on a small network, and Crowley et al. (2025) found that, beyond about 40 sensors on the English network, their placement held no more information than simply choosing the highest-flow nodes. Redundancy is therefore a property a placement objective must penalise, not merely a by-product. Second, mutual information is a general-purpose objective that is not tied to any single service, which is its stated advantage, but Cheng et al. (2026) show that a placement optimised for general information is not optimal for a specific prediction. The six formulations are therefore best read not as competing algorithms for one problem but as answers to different questions: where contaminants would be detected (Simone et al.), where blockages would be seen before overflow (Ninh et al.), where flow could be reconstructed (Ding et al.), where the network state is best estimated (Crowley et al.; Wang et al.), and where a stated prediction is best informed (Cheng et al.).

Scale separates them further. The deterministic solver handled 531 nodes in under a second because hydraulics were precomputed; the genetic algorithm handled some 100 to 190 candidate maintenance holes; the modified greedy algorithm took about 14 hours for 250 sensors on 1,020 nodes; and Bayesian optimal design needed a trained surrogate to rank 2,576 candidates for a single sensor. For a network of SA Water's size, the greedy route with precomputed simulation statistics is the realistic path, with Bayesian design best cited as the justification for what the greedy objective should measure.

3 Detecting Blockages and Overflows from Sensor Data

A placement formulation decides where sensors go; a detection rule decides what each placed sensor can see. The three sources in this section show that the second choice changes the first.

3.1 Utility practice at Sydney Water

Drenoyanis and Prackwieser (2022) describe how Sydney Water, an Australian utility directly comparable to SA Water, detects dry-weather blockages, most of which occur in small-diameter reticulation pipes and are caused by tree roots, wet wipes, fat, oil and grease, or a combination. Three device types are in use: float switches or level switches that alarm at a set level, ultrasonic or radar level sensors that measure every 15 minutes and report once a day or on alarm, and flow sensors. Level sensors are preferred where the consequence of failure is very high or the chamber is very shallow, because the rate of rise lets operators estimate how long they have before an overflow.

Site selection is a ten-step, risk-based process rather than an optimisation: prioritise a sub-catchment by its dry-weather overflow history and licence limits; inspect and repair on site, including signal testing; rate each asset by the likelihood of blocking and the environmental consequence of failure; combine ratings along pipeline chains as a probability of failure in series; have subject-matter experts assess sites spatially; and check safety, access, heritage and traf-

fic requirements. Device specifications are equally practical: corrosion-resistant casings, IP68 sealing, vermin-resistant cables, external antennas at about 90% of sites, long battery life, and NB-IoT preferred over LoRaWAN for metropolitan use. Three analytics methods raise alarms: comparing pumping-station inflow against historical diurnal patterns, which localises a fault only to potentially tens of kilometres of network; comparing upstream and downstream chambers, where a sustained positive normalised flow error indicates a choke downstream and a negative one indicates a choke upstream; and correlating digital water-meter consumption with derived sewer flow, which held at 0.97 on a dry day but fell to 0.79 on a rainy one. The programme detects on average 20 blockages a month, had cleared more than 300 at environmentally high-risk sites, and is credited with about $400,000 a month in avoided incident costs. Many of the chokes found were in bushland, where an overflow could have run unseen for days.

The paper's limitations are stated plainly. Because devices report once a day to preserve battery, comparing two chambers in real time is not currently possible; inflows entering between two devices must be predicted from history; and no detection rate or false-alarm rate against a ground-truth set of blockages is reported. It is a non-peer-reviewed industry paper.

3.2 Level-based identification and sensor spacing

Li et al. (2023) formalise the paired-chamber idea. After a blockage, level rises upstream and falls downstream, so a double-threshold rule on the rate and the range of level change can declare a blockage downstream of a gauge (both above positive thresholds) or upstream of it (both below negative thresholds). Thresholds were set from pilot-scale experiments in DN200 pipe and full-scale experiments with six level sensors in DN300 rural gravity sewers in China, with artificial blockages of 0.1 to 0.2 kg dry weight.

Within a minute of blocking, a level rise was detectable within 4 m upstream, and within four minutes the level peaked at 300% to 700% above its prior value. Identification within 4 m upstream reached 95.3%, and the overall success rate of the discriminant was 80.4% at pilot scale and 95.6% at full scale. Although a change could still be detected 180 m from a blockage, the authors advise that monitoring points should be no more than 100 m apart if identification performance is to hold. This is the only source reviewed that gives an empirical upper bound on sensor spacing for small-diameter gravity sewers. Its thresholds, however, are tied to the rigs, diameters and assumed 40 L per person per day used, the pilot rig used clean water, and 16.7% to 20% of blockage cases were missed.

3.3 Acoustic localisation within a pipe

Ninh et al. (2026), from the same Adelaide group as the coverage formulation above, test a complementary technology. Two microphones 0.5 m apart record sound travelling along the pipe, excited by white noise or by ambient noise, and a paired impulse response function turns a blockage into a pair of opposite-sign spikes whose timing gives its location and whose size gives its severity. The paper motivates the work with local figures: more than 3,500 blockages from "unflushable" items in the metropolitan Adelaide network in 2022–2023, and more than two million dollars spent by SA Water in 2024 removing such material. It also notes that level sensors struggle with foaming or vapour-producing flows and at complex junctions with several inflows.

In an 11.5 m laboratory rig of 114 mm PVC pipe at 4° slope, the method located solid and tree-root blockages covering 37%, 54% and 82% of the pipe area with location errors below 1% and estimated blocked area with errors below 8%, with and without flow. The paired spikes remained visible even with no deliberate excitation, so ambient pipe noise could serve as the source, although an excitation above about 25% of the test volume is recommended for robust detection. The authors note that the rig is far shorter than the distance between real maintenance holes and that complex blockage geometry can produce ambiguous signatures, so

field testing is still required.

3.4 Implications for placement

The three detection studies converge on one point the placement formulations treat as settled: the region a sensor covers is not a fixed property of its location. Field data in Ninh et al. (2025) show a level sensor reliably seeing blockages downstream of it but not upstream, and the coverage formulation is built on that asymmetry; Li et al. (2023) detect in both directions in controlled experiments; and Sydney Water infers the side of a choke from the sign of a flow error between two chambers. A placement objective for the project must therefore state the detection rule it assumes, and should be tested against the 100 m spacing bound of Li et al. (2023) and the steep-terrain limitation of Ninh et al. (2025). Acoustic sensing does not compete with level sensing for the same role: it localises a blockage within a reach after a level alarm or during inspection, rather than providing continuous network-wide coverage.

4 Factors Influencing Sewer Condition

Malek Mohammadi et al. (2020) review condition prediction models published between 2001 and 2019 and catalogue the variables used in them. Following Davies et al., the factors are grouped as physical (age, material, diameter, length, depth, slope, shape, joint type, invert elevations), environmental (soil type, corrosivity, moisture, groundwater level, bedding and backfill, road type, traffic, root interference) and operational (blockages, flow velocity, sediment level, infiltration and exfiltration, sewer type, maintenance history). Of the twenty independent variables surveyed, pipe age, diameter and length emerge as the most frequently significant contributors to deterioration.

Three observations from this review carry into the parameter catalogue (D2).

4.1 Several factors are contested rather than settled

Diameter, depth, slope and pipe length all appear with contradictory signs across studies. Larger diameters are associated with both lower risk (attributed to more careful design and installation of larger assets) and higher risk (attributed to greater surface area in contact with sewage and soil). Depth is insignificant in the majority of the models reviewed. The review's own explanation is that condition models are fitted to local geography, so a factor's direction of effect is not transferable between utilities. For this project, the implication is that the catalogue should record each parameter's mechanism and its evidential status, not a single assumed sign.

4.2 Data availability biases which factors are studied

Physical attributes dominate the literature because they are cheap to collect and already held in asset inventories; environmental and operational data are frequently absent. The review recommends greater use of environmental and operational variables, specifically soil type, bedding material, flow rate and soil corrosivity. This mirrors the data-availability constraint the project itself faces and supports documenting availability status explicitly for each catalogued parameter.

4.3 Condition is assessed against defined rating protocols

The review distinguishes structural condition (defects, strength, resistance to external load) from operational condition (ability to meet service requirements), and notes that structural results drive rehabilitation decisions whereas operational results drive cleaning and maintenance. Rating protocols include WRc, PACP, the NRC protocol and, relevant to an Australian deployment, the WSAA standard, with condition typically graded from 1 to 5.

4.4 A larger and more recent evidence base

Ma et al. (2026) extend this picture with a systematic review of 71 peer-reviewed studies published between 2016 and 2025 on AI-driven sewer condition assessment, roughly triple the span covered by Malek Mohammadi et al. (2020). Restricting attention to the subset of that review using structured, non-visual data, that is, the traditional machine-learning studies most comparable to condition-prediction practice, pipe age and diameter recur as the two most frequently identified influential factors, cited in 54.2% and 25.0% of the reviewed studies respectively, followed by pipe material, length and slope; this corroborates, at a larger sample size, both the age and diameter findings of Malek Mohammadi et al. (2020) and the mutual-information ranking reported by Ma et al. (2025). The review also confirms the data-availability bias already noted above: physical attributes appear in every study surveyed, geospatial and socio-environmental factors in roughly half, and hydraulic or chemical properties, the category closest to the flow and level behaviour this project needs to monitor, in under one fifth. For the parameter catalogue (D2), this strengthens the case for treating pipe age, diameter and material as a well-evidenced core, while confirming that growth-related load and flow behaviour remain comparatively understudied and therefore require the project's own analysis rather than a literature-derived weighting.

Two further points from Ma et al. (2026) carry into the placement method (D3) and the explainability requirement (R4). First, the review's summary of standardised inspection protocols, PACP, MSCC, WSA 05, HKCCEC, CSA PLUS 4012–10 and EN 13508–2 among others, converges on a five-point condition scale in which grade 4 marks the threshold for required intervention and grade 5 marks imminent or actual failure; this scale is a candidate reference point for expressing the project's own placement priority in familiar, protocol-aligned terms when presenting recommendations to the asset management team, rather than in raw topological or observability units that carry no meaning outside the project. Second, the review's discussion of interpretability, in particular SHapley Additive exPlanations (SHAP) for quantifying factor contributions and Bayesian or Monte Carlo dropout methods for attaching confidence intervals to a prediction, is offered there as a remedy for the low transparency of machine-learning condition models; the same techniques are a plausible route for the project to attach a defensible confidence statement to a placement recommendation once the parameter catalogue (D2) is finalised, directly supporting the explainability requirement (R4) that any recommendation must be understandable to a non-specialist approver.

5 Quantifying Factor Influence: Bayesian Network Modelling

Ma et al. (2025) address the disagreement identified above by estimating factor influence from data rather than from domain assumption. Fourteen influence factors, spanning pipe physical properties, environmental context and climate, were assembled into an integrated database for the Hong Kong sewer network, which spans approximately 4,700 km. CCTV defect records were joined to GIS pipe records on pipe identifier, and environmental and climate attributes were attached by spatial join; approximately 23,000 pipe records remained after cleansing.

The target variable is an aggregated defect-based condition on a six-level scale, derived by taking the maximum of the internal condition grade, the service condition grade, and a structural performance grade that adjusts the internal grade for installation age and slope location. Network structure was learned by a score-based Bayesian search using the Bayesian Information Criterion, producing a directed acyclic graph of 15 nodes and 31 links in which pipe condition is the child of pipe age, pipe diameter and population.

Factor influence was then ranked by three sensitivity measures: mutual information, joint probability, and true risk influence, the last computed by forcing each state of a factor to certainty and taking the mean of the resulting highest and lowest deviations from the prior. All three agree on the leading factors. By mutual information, pipe age ranks first (0.124), followed

by diameter (0.032), population (0.027) and soil type (0.024); every remaining factor scores below 0.02. Scenario analysis identified the combination most associated with critical condition as age above 50 years, diameter below 200 mm, population density below 15,000, and fill or granitic soils; forcing this combination raised the probability of critical condition from 17% to 72%. Validation against a withheld inspection record returned the correct condition class with 85% probability.

Two aspects of the method are transferable regardless of whether the numerical findings are. First, the Bayesian network handles missing data and supports both forward and backward inference, which suits a setting where some network attributes are known to be incomplete. Second, the sensitivity measures provide a defensible basis for reducing a long candidate parameter list to a working subset, which is precisely the task PO2 sets.

6 Inflow and Infiltration as a Network-Screening Signal

Inflow and infiltration (I/I) is both an indicator of structural condition and a source of the extra load that causes surcharge in wet weather. Two sources show that it can be identified from monitoring a utility already has, which makes it a candidate input to the placement parameter catalogue (D2) and a way to rank sub-catchments before any new sensor is installed.

6.1 Machine learning for rainfall-derived inflow and infiltration

Wang et al. (2024) predict sewer flow with random forest and long short-term memory (LSTM) models at two sites in Ningbo, China, one combined (Bailong) and one separate (Wantou), and take rainfall-derived inflow and infiltration (RDII) as the difference between observed wet-weather flow and a monthly-average dry-weather pattern. The LSTM performed better, and maximum relative errors on RDII volume across eight rainfall events (about 9.5 to 78.5 mm) were under 7%. Two results matter for the project. First, accuracy fell steadily as the data interval lengthened from 5 to 60 minutes, by 0.15 to 0.23 in R^2 depending on site and model, which sets a sampling requirement for any sensor whose data feed this kind of analysis. Second, flow in the separate system took two to three days to return to dry-weather levels after rain, against a quick return in the combined system, which the authors read as evidence of mixed connections or structural defects. A long recovery tail is therefore a cheap indicator of a defective separate sewer. The RDII target is itself derived from an assumed dry-weather pattern, and only two catchments in one city were studied.

6.2 Sub-catchment screening from pumping-station data

Zhang et al. (2020) compare an adaptive neuro-fuzzy inference system and a multilayer perceptron for identifying I/I in the sub-catchments of two pumping stations in Espoo, Finland, a network of more than 900 km and about 200 pumping stations. Because the city has only two rain gauges, rainfall came from weather radar, which covered every catchment. Flow was split into dry- and wet-weather sets by rainfall thresholds, and infiltration was identified where the normalised minimum dry-weather flow stayed above zero, meaning night-time flow exceeded what water use alone would produce. Both models identified I/I in both sub-catchments; the neuro-fuzzy model performed better overall, and wet-weather results were sensitive to the threshold because few wet-weather data points remained. The authors state that the flow data were not yet sufficient to quantify I/I in all situations, and only two stations from about 200 were studied.

6.3 Relevance to placement

Neither study places sensors, and neither should be read as placement evidence. Together, however, they give two independent indicators that can be computed from existing pumping-station or SCADA records: minimum night flow above the water-use baseline, and slow recovery to dry-weather flow after rain. Both identify where extra load is entering the network, which is the same quantity that growth adds, so a sub-catchment ranking built from them is a defensible first screen for where to concentrate monitoring. The 5-minute sampling result also has to be reconciled with the 15-minute, once-a-day reporting that keeps utility devices running on battery (Drenoyanis and Prackwieser, 2022); this tension is raised in Section 8.7.

7 Sensor Technology Performance and Data Value

Every placement formulation in Section 2 treats a sensor reading as dependable once a node is instrumented: the placement problem is where to put sensors, not whether a sensor at a given node can be trusted, nor whether a cheaper device would do. The partial exceptions are the noise and dropout tests of Ding et al. (2026) and the additive-noise model of Crowley et al. (2025), both of which assume the same noise at every node. The sources in this section test those assumptions directly.

Alshami et al. (2023) installed ultrasonic flow sensors and ultrasonic level sensors together in 12 maintenance holes across four districts of the Hong Kong sewer network, spanning inlet pipe diameters of 0.15–0.6 m, and monitored them continuously for one year (June 2019 to June 2020), yielding approximately half a million observations.

7.1 Reliability findings

The flow sensors performed poorly against two independent checks. The correlation coefficient between flow-sensor velocity and level-sensor depth across all maintenance holes was 0.36, and it varied widely by location, from 0.92 at one maintenance hole to -0.13 at another. Flow rates computed from the flow sensors were compared against flow rates computed from level-sensor depth via Manning's equation, and the average relative error across all maintenance holes was 72.14%, with individual maintenance holes ranging from 19% to 97.6%. At one maintenance hole, the flow sensor recorded velocities exceeding 30 m/s while the level remained essentially constant, an implausible reading for a gravity sewer that the authors attribute to sensor malfunction under a specific combination of conditions rather than to genuine flow.

7.2 Causes identified

Three design and operational conditions were found to degrade flow-sensor accuracy, each traced by comparing maintenance holes that performed well against maintenance holes that did not.

Outlet pipe diameter. The ultrasonic flow sensors measure velocity by firing a pulse across the flow surface and timing its reflection; this depends on a consistent flow direction. In small-diameter outlet pipes the flow direction becomes inconsistent as sewage exits the maintenance hole, and the smaller the pipe, the more pronounced the effect. Two maintenance holes with otherwise identical design (single inlet and outlet) but outlet diameters of 0.15 m and 0.225 m showed markedly different reliability, with the narrower pipe performing worse.

Sewage level relative to the outlet pipe. Once the sewage level rises above the outlet pipe's cover level, the dominant motion at the sensor becomes vertical rather than horizontal, which the horizontal-flow sensor cannot resolve. Readings in this regime were either near zero or erratic, including the implausible high-velocity readings noted above. Because sewage level

can exceed the outlet cover level from ordinary high usage as well as from a blockage, this failure mode also produces false positives for blockage and overflow detection specifically at the moments when detection matters most.

Multiple inlet and outlet pipes with varying invert levels. Ten of the twelve maintenance holes studied had two or three inlet pipes. Where a secondary inlet contributed materially to inflow, and particularly where inlet pipes had different downstream invert levels, flow direction at the sensor became inconsistent and accuracy fell. Where a secondary inlet's contribution was small relative to the main pipe, accuracy was largely unaffected, indicating that inlet count alone is not the determining factor; relative inflow contribution and invert-level variation are.

7.3 Proposed alternative and its relevance

Because level sensors alone remained reliable throughout, the authors propose deriving a vertical velocity from successive level readings (five minutes apart in the study) rather than relying on the horizontal flow sensor. A rising level with consistently positive vertical velocity signals a blockage; the magnitude and sign of the vertical velocity, combined with maintenance hole geometry (cover level, invert level, pipe diameter), allow a simple estimate of remaining time to overflow. Validated against the same twelve maintenance holes, this measure tracked the observed sewage level far more closely than the flow sensors did, and requires only the cheaper of the two sensor types.

Three implications follow for this project. First, the reliability of a sensor at a candidate node is not uniform across the network: it depends on outlet diameter, on how often level is expected to exceed the outlet cover (itself a function of the growth-driven load the project is designed to address), and on the number and configuration of connected pipes at that maintenance hole. Second, the finding motivates why level sensors, not flow sensors, are the more dependable instrument in exactly the small-diameter, multi-inlet conditions common at the periphery of a growing network, which is also where the project's target overflow mechanism concentrates. Third, the vertical-velocity construction is a candidate observable for the network visibility measure the project has not yet defined: unlike centrality, which is purely topological, it is a directly measurable quantity tied to the physical event (rising level) the project needs to detect.

7.4 Corroboration from adjacent inspection technologies

Ma et al. (2026) broaden this picture beyond flow and level sensors. Reviewing laser profiling and LiDAR as alternatives to CCTV for internal sewer inspection, they note that both methods offer higher data resolution and acquisition efficiency than CCTV but remain susceptible to environmental interference, point-cloud noise from surface reflection or debris, and, notably, a reliable operating range limited to above the water level, the same operating boundary that Alshami et al. (2023) identify as the point at which flow-sensor accuracy for the twelve Hong Kong maintenance holes collapses. That the same limitation recurs across an entirely different sensing modality strengthens rather than merely repeats the reliability finding above: any technology that measures at or below the design water surface, whether by ultrasonic flow, laser range-finding or LiDAR, degrades once the surface itself is disturbed by a rising or turbulent flow, which is precisely the operating condition the project's overcapacity application must observe most closely.

7.5 Low-cost prototype hardware

Utepov et al. (2023) built and bench-tested a low-power IoT chamber monitor in Astana, Kazakhstan: a sensing unit in the chamber with an upward-facing sensor for the manhole cover, a downward-facing sensor for the wastewater level and an infrared module on the outlet pipe for

blockage, linked by 433 MHz LoRa to a data-gathering station with a reported range of up to 10 km, and a web GIS that shows colour-coded level, cover and blockage alerts. Of four distance sensors tested over simulated chamber distances, a laser time-of-flight sensor was most accurate over 0.2–1.0 m (suited to detecting the cover) and a LiDAR sensor was most accurate over 1.0–5.0 m, staying within about ±8 cm to 5 m (suited to the level). The low-cost ultrasonic sensor tested gave the poorest results, with readings stuck at 0.22–0.23 m for targets 2–5 m away. Materials for one station and one sensing unit cost about US$170. The tests were bench simulations rather than field trials, and the authors list real-sewer validation and a cost–benefit analysis as future work. For the project, the result is a caution rather than a recommendation: the cheapest ranging sensors can fail silently by reporting a plausible constant value, so device choice and a check for stuck readings belong in the reliability layer of the placement method.

7.6 Soft sensors: deriving flow from level

Lin et al. (2025a) train a multilayer perceptron to estimate flow at a manhole from a window of level readings around the target time. A window is needed because the depth–flow relationship in a manhole is not a rating curve: free-surface and pressurised, sub- and supercritical flows make it multivalued, and both earlier and later depths carry information about the current flow. Trained on a year of 1-minute level and Doppler flow data at three hydraulically different manholes in China, plus a CFD-simulated manhole, the model gave satisfactory estimates at all four; a window of about 4 minutes and 1-minute sampling were sufficient, and on the simulated manhole the mean relative error stayed between 2.9% and 3.4% under added noise. The constraint is that every manhole needs its own flow campaign to train the model, and a bias in the training flow data was amplified in the output (26% input error became 38%).

Lin et al. (2025b) reduce that constraint by reading level at two adjacent manholes and running the estimate on a Raspberry Pi linked by LoRa. A Saint-Venant model with only roughness and sediment depth calibrated reached a mean relative error of 14.7% after three hours of training data and 13.0% after one day, while a neural network was more accurate when data were plentiful (5.1% after 35 days) but needed wet-weather events to learn. A physics-informed neural network failed outright. The Saint-Venant sensor also recovered roughness and sediment depth, and an accumulating sediment depth is a precursor to blockage. Both real pipes, however, already had flow meters at both ends, so the result shows that less flow data is needed, not that none is.

For the placement problem, the pair changes the cost model rather than the location model. If a level sensor plus a short commissioning visit can stand in for a flow meter, the budget constraint becomes how many level sensors can be afforded, which raises achievable density by a large factor, and the natural unit of instrumentation for flow becomes a pipe reach between two manholes rather than a single node. Because the estimate uses future as well as past levels, it is slightly retrospective, which suits modelling and I/I accounting better than real-time alarms.

7.7 The information content of binary sensors

Wani et al. (2015) ask how much a yes/no sensor, such as a float switch or an overflow detector, is worth for calibrating a drainage model. They build a formal likelihood for binary observations, treating a signal as the event that the true response (a SWMM simulation plus an autocorrelated error term) exceeds a threshold, and infer model parameters by Bayesian sampling. On the Adliswil catchment near Zurich (458 junctions), calibration on binary data derived from the outlet flow reached a Nash–Sutcliffe efficiency of 0.78 against 0.80 with continuous flow, and narrowed parameter ranges well beyond the priors. Information peaked when the threshold sat in the middle of the range the system actually experiences and fell towards zero when the switch almost never or almost always triggered. The authors conclude that an optimal design would combine many cheap sensors with a few accurate flowmeters. The paper is a seven-page

conference extended abstract, the binary data were synthetic, sensors were assumed error-free and only the outlet was used, so it supports the principle and the threshold finding rather than precise numbers.

For the project this has two consequences. First, a float switch is not a degraded level sensor but a sensor with a different information profile, and its value depends on where its threshold sits relative to the depths seen at that particular chamber, so alarm levels should be set per chamber rather than network-wide. Second, together with Sydney Water's mix of switches, level sensors and a few flow meters, it reframes the placement question from how many locations to which mix of sensor types, and where each type goes.

8 Synthesis and Gaps

Table 1 classifies all nineteen sources against the PO1 scheme, grouped as in Section 1. Eight gaps follow from this set of sources.

Table 1: Classification of reviewed sources against the PO1 scheme.

Source	Formulation	Objective function	Input parameters	Network type	Purpose	Applicability to trunk sewer
Placement formulations						
Simone et al. (2023)	(a) MINLP, global solver; (b) centrality ranking	(a) Maximise detection reliability over injection scenarios; (b) none, ranking only	(a) Topology, flow velocity, concentration (SWMM-derived), detection threshold; (b) topology, nodal inflow, slope	Gravity sewer, directed graph; 77-node benchmark and 531-node real network	Contaminant / illicit intrusion detection	High: directed-graph abstraction and decoupled hydraulics both transfer
Ninh et al. (2025)	Blockage detection condition on invert elevation and typical depth; breadth-first search; genetic algorithm	(i) Minimise sensors for full coverage; (ii) maximise pipe length covered by a fixed number per investment phase	Node inverts, maintenance-hole depths, connectivity and flow direction, typical depth, alarm threshold (70% of depth)	Two real gravity sanitary sewers (location not stated), 103 and 189 maintenance holes, 150 mm pipes	Dry-weather blockage detection before overflow	High: Adelaide research context, small reticulation pipes and level sensors as in SA Water's network; no hydraulic model needed; typical depth is a fixed input, so growth is not represented
Ding et al. (2026)	Data-driven sparse sensing: SVD for a reduced temporal basis, QR factorisation with column pivoting for sensor selection	Maximise system-level flow-reconstruction accuracy (NSE) against held-out SWMM-simulated events	SWMM-simulated flow series across an ensemble of rainfall and hydrologic scenarios	Storm sewer, 77-node network, Duluth, Minnesota	System-wide flow reconstruction for flood monitoring	Moderate: offline-basis architecture and reconstruction-NSE measure transfer; linear framework is weak at threshold-driven overflow nodes
Crowley et al. (2025)	Mutual-information maximisation under a Gaussian state model; one-step modified greedy algorithm	Maximise mutual information between all node states and sensor observations	Covariance of simulated flow or depth at every node; sensor noise variance	479-node foul network (England); 1,020-node combined network (Bellinge, Denmark)	General state estimation for network monitoring	High: scales beyond 1,000 nodes and can extend an existing network; needs a calibrated hydrodynamic model; assumes equal noise at every node
Wang et al. (2023)	K-means clustering of node attributes, then re-clustering by closeness centrality or node inflow	None explicit; layouts scored by joint entropy (max) and total correlation (min)	SWMM level and COD series, elevation, coordinates; centrality or dry-weather inflow	31-node combined (SWMM Example 8); 786-node separate industrial sewer, Wenzhou	Routine monitoring	Moderate–high: reproducible baseline; inflow-weighted variant maps to growth; no test of actual fault detection
Cheng et al. (2026)	Bayesian optimal experimental design with a neural-network surrogate	Maximise expected information gain about parameters or a stated prediction	Six-parameter prior, one storm, 2,576 candidates, public-data feasibility screen	Urban overland flood domain, Detroit; not in-pipe	Flood prediction	Low numerically, high methodologically: objective-first workflow and feasibility screen transfer
Blockage and overflow detection						

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Table 1 continued

Source	Formulation	Objective function	Input parameters	Network type	Purpose	Applicability to trunk sewer
Drenoyanis & Prackwieser (2022)	Practitioner survey; ten-step risk-based site selection	None formal; likelihood × consequence rating, with probability of failure along series chains	Overflow history, licence limits, pipe size and location, access, safety, signal tests	Sydney Water reticulation network	Dry-weather blockage and overflow detection	High: Australian utility practice; supplies a constraints checklist and a value case
Li et al. (2023)	Double-threshold rule on rate and range of level change	Not applicable; reports identification rate	Level series up- and downstream of a blockage	Rural gravity sewers DN150–DN300 (China), pilot and full scale	Blockage identification and relative positioning	Moderate: mechanism and 100 m spacing bound transfer; thresholds do not
Ninh et al. (2026)	Paired impulse response function from two microphones	Not applicable; reports location and blocked-area error	Acoustic pressure at two points 0.5 m apart; white-noise or ambient excitation	11.5 m laboratory PVC pipe, 114 mm	Locate and size blockages within a reach	Low–moderate: laboratory only; complements level sensing for localisation rather than coverage
Network attributes: condition and inflow/infiltration						
Malek Mohammadi et al. (2020)	Narrative review; no optimisation	Not applicable	20 physical, environmental and operational attributes	Gravity sanitary, storm and combined sewers	Structural and operational deterioration	Moderate: parameter taxonomy transfers; effect directions do not
Ma et al. (2025)	Bayesian network, score-based structure learning	Posterior probability of defect-based condition class	14 factors: physical, environmental, climate; CCTV defect records	Municipal sewer network, approx. 4,700 km (Hong Kong)	Defect-based condition and inspection prioritisation	Moderate: method transfers; coefficients are specific to Hong Kong
Ma et al. (2026)	Systematic review (71 studies) of traditional ML and deep-learning models	Not applicable; reports performance of the models surveyed	ML: physical, geospatial, socio-environmental, operational, hydraulic and climatic factors; DL: CCTV, laser, LiDAR	Aggregated across 71 studies, many countries	Condition prediction (ML) and CCTV defect analysis (DL)	Moderate: factor taxonomy and age/diameter evidence transfer to D2; does not address placement
Wang et al. (2024)	Random forest and LSTM flow prediction; RDII as wet minus dry-weather flow	Prediction accuracy; RDII volume error	Flow and rainfall at 1 min, resampled to 5 min; engineered features	Combined (Bailong) and separate (Wantou) sewers, Ningbo	RDII quantification	Moderate: sampling-interval and recovery-time evidence transfer; not placement
Zhang et al. (2020)	Neuro-fuzzy and multilayer perceptron flow models split by rainfall threshold	RMSE and R^2 of flow models	Hourly pumping-station SCADA flow; weather-radar rainfall	Two pumping-station sub-catchments, Espoo	Sub-catchment I/I identification	Moderate: minimum-night-flow screen and radar substitution transfer
Sensor technology and data value						

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Table 1 continued

Source	Formulation	Objective function	Input parameters	Network type	Purpose	Applicability to trunk sewer
Alshami et al. (2023)	Empirical field study; no optimisation or prediction model	Not applicable; reports correlation and relative error against Manning's equation	Maintenance-hole design (pipe diameters, invert levels, inlet/outlet count); paired flow and level readings	Small- and medium-diameter gravity sewer, 12 maintenance holes, Hong Kong	Sensor reliability for blockage and overflow detection	High: diameter range and maintenance-hole configurations overlap with SA Water's network
Utepov et al. (2023)	IoT prototype; bench tests of four distance sensors	Not applicable; reports measurement deviation	Distances 0.2–5.0 m; outlet infrared blockage module	Simulated sewer chamber (Astana)	Level, manhole-cover and outlet-blockage alerts	Low–moderate: sensor-choice evidence and cost; not field-tested
Lin et al. (2025a)	Multilayer perceptron soft sensor on a window of level readings at one manhole	Minimise flow-estimation error	Level and Doppler flow at 1 min for training	Three manholes (China) and one CFD manhole	Flow from level	Moderate: lowers cost per flow point; needs a flow campaign per site
Lin et al. (2025b)	Saint-Venant, MLP and PINN soft sensors on two adjacent levels, run on-device	Minimise flow-estimation error	Two level series; hours to days of flow data	600 mm and 1,200 mm pipes, 70 m and 113 m reaches (China)	On-device flow monitoring	Moderate: reach-based unit of instrumentation; short commissioning
Wani et al. (2015)	Bayesian calibration with a likelihood for binary observations; MCMC	Posterior parameter estimates; NSE	Binary threshold exceedance at the outlet (derived from flow); rain gauges	Adliswil catchment, 458 junctions (Switzerland)	Model calibration	Moderate: threshold rule for level switches; not placement; conference abstract

8.1 Objective mismatch

Most of the condition- and placement-oriented studies reviewed address contaminant detection, deterioration, general flow reconstruction or general state estimation. Ninh et al. (2025) is the exception that comes closest: its objective is blockage detection before overflow in gravity sanitary sewers, and it comes from the Adelaide group studying South Australia's smart sewer programme. Even there, the objective is the length of pipe a sensor would see under today's typical depths, not whether a rising level caused by increased connected load would be detected in time to act. Ding et al. (2026) reconstructs a genuine hydraulic quantity but, as a linear reduced-order method, is explicitly weakest at the threshold-crossing behaviour that defines an overflow event, and Crowley et al. (2025) and Wang et al. (2023) maximise information about the network in general rather than about any one event. Cheng et al. (2026) explain why this matters: a placement is optimal only for the prediction it encodes. The placement formulations reviewed are reusable, but their objective functions only partly are: an observability measure for rising sewer level under increased connected load must be defined rather than adopted, and the coverage condition of Ninh et al. (2025) is the most natural starting point for it.

8.2 Head nodes and upstream risk

The topological metric in Simone et al. (2023) assigns null relevance to head nodes and therefore excludes them. For contaminant detection this is correct, since a head node receives nothing. For growth-driven overcapacity it may be the wrong exclusion, because new residential connections attach at the periphery of the network and the resulting surcharge can originate upstream. Ninh et al. (2025) has a similar blind spot by design: because a sensor covers only pipes downstream of it, the most upstream nodes cannot be covered at all, leaving 43 of 267 nodes uncovered in the flat system and 97 of 323 in the steep one. Any centrality- or coverage-based objective adopted for this project must be tested against the case of surcharge originating at the network's growing edge before use.

8.3 Load representation

Both Malek Mohammadi et al. (2020) and Ma et al. (2025) treat population or population density as a static environmental attribute of a pipe. The placement formulations do the same in different forms: Wang et al. (2023) weight sensors by today's dry-weather inflow, Crowley et al. (2025) by covariances estimated from today's simulated flows, and Ninh et al. (2025) by today's typical depth H_s. The project requires the opposite framing: load as a quantity that changes when a development is connected. In the coverage condition of Ninh et al. (2025), a higher connected load raises H_s and therefore shrinks the set of nodes each sensor covers, so a layout that gives full coverage today may not after growth. No source reviewed models placement under a changing load distribution, and this remains the sharpest gap in the reviewed set and the clearest opening for the project's contribution.

8.4 Absence of a common evaluation measure

The sources now evaluate placement against at least seven different measures: detection reliability (Simone et al., 2023), covered pipe length (Ninh et al., 2025), reconstruction NSE (Ding et al., 2026), NMSE of unmonitored nodes and mutual information (Crowley et al., 2025), joint entropy and total correlation (Wang et al., 2023), expected information gain (Cheng et al., 2026) and, for condition models, posterior probability or rating accuracy (Ma et al., 2025; Malek Mohammadi et al., 2020). None is a measure of overflow observability under growth-driven load. NSE and NMSE measure how well an unobserved series can be reconstructed in general, not whether a rising level relative to a specific overflow threshold would be detected in time to act; covered pipe length assumes that detection follows from geometry; and the information measures

are, by design, independent of any specific event. Only Li et al. (2023) report a detection rate against actual blockages, and only in controlled experiments. This is consistent with the absence of a unified quantitative evaluation process noted elsewhere in the sensor placement literature, and it reinforces the operational definition of network visibility as the project's principal open question with the client. The project's own blockage and growth simulations can supply what the literature lacks: a test of every candidate layout against simulated fault events.

8.5 Uniform sensor reliability is assumed, not tested

Both formulations in Simone et al. (2023), and by extension any placement method built the same way, treat detection at an instrumented node as reliable once installed; the optimisation and the centrality ranking differ only on where to place sensors, not on whether a given node can be trusted to report correctly. Crowley et al. (2025) go further than most by modelling sensor noise, but assume the same noise at every node. Alshami et al. (2023) show this assumption fails in small-diameter, multi-inlet maintenance holes, exactly the configuration expected at the periphery of the network where growth-driven connections attach; Utepov et al. (2023) show that a low-cost ranging sensor can report a plausible constant value regardless of the target; and Lin et al. (2025a) show that bias in soft-sensor training data is amplified rather than averaged out. A placement method for this project should therefore weight candidate nodes by expected in-situ reliability, informed by outlet diameter, inlet configuration, and expected level relative to the outlet cover, rather than by topological or hydraulic importance alone. This also bears on sensor type selection (R8): the evidence favours a level-sensor-led design over a flow-sensor-led one specifically in the small-diameter, multi-inlet locations most likely to host new connections. Ding et al. (2026) is a partial exception in that it tests robustness to measurement noise and to sensor dropout directly, but both are properties of the reconstruction algorithm's tolerance to missing or noisy data, not of the physical sensor malfunction modes Alshami et al. (2023) document; a placement method for this project should therefore weight candidates by both kinds of reliability, algorithmic robustness to missing readings and physical robustness of the reading itself, rather than treating either as a stand-in for the other.

8.6 No standard link to the condition-grade scale asset managers already use

Ma et al. (2026) show that a five-point structural/operational condition scale, with intervention triggered at grade 4 and failure at grade 5, is the practitioner-facing convention across the protocols it surveys (PACP, MSCC, HKCCEC and others). None of the placement formulations reviewed here express their output in that scale, nor do the condition-modelling sources Malek Mohammadi et al. (2020) and Ma et al. (2025). Presenting a placement recommendation (D3) or business case (D7) in a topological score, an NSE value, a covered length, a mutual-information gain or a posterior probability risks the same approval-stage failure a non-specialist funding approver would reject outright; a translation from whatever observability measure the project adopts into a grade-4-equivalent priority threshold is worth defining early, not left to the write-up. Sydney Water's reported value case, about 20 blockages detected a month and $400,000 a month in avoided costs (Drenoyanis and Prackwieser, 2022), shows the form such a translation can take.

8.7 Telemetry and deployability sit outside the formulations

Every formulation reviewed assumes that any candidate node can host a sensor and that its data arrive when needed. Utility practice contradicts both. Sydney Water devices measure every 15 minutes and report once a day to preserve battery, so a real-time comparison between two chambers is not currently possible, and about 90% of sites need an external antenna (Drenoyanis and Prackwieser, 2022); Ninh et al. (2025) restrict candidates to maintenance holes on safety

grounds; and Wang et al. (2024) show that analytics lose accuracy as the sampling interval lengthens beyond 5 minutes. Cheng et al. (2026) is the only source that builds a feasibility screen into the design process. A placement method for this project should therefore apply a deployability screen (access, safety, communications, power) before optimisation and state the sampling and reporting interval each detection rule needs, because a layout that depends on synchronised, high-frequency readings from adjacent chambers is not deployable as designed.

8.8 One sensor type is assumed everywhere

Each placement formulation places one kind of sensor. The technology sources show that the real choice is a mix: Sydney Water uses float switches widely and level sensors where consequence is highest (Drenoyanis and Prackwieser, 2022); Ninh et al. (2025) describe ultrasonic level sensors reading every 5 minutes alongside float switches with low (150 mm) and high (300 mm) alarms; Wani et al. (2015) show that binary switches carry substantial information if their thresholds are set well; Lin et al. (2025a, 2025b) show that level sensors can also deliver flow; and Ninh et al. (2026) add acoustic localisation within a reach. Placing a fixed budget across sensor types with different costs, information profiles and failure modes is a different optimisation from placing identical sensors, and none of the sources attempts it. This is the second clear opening for the project's contribution, after load representation.

9 Conclusion

The nineteen sources reviewed support four conclusions for the project.

First, the method for placing sensors is largely available; the objective is not. Six placement formulations were reviewed, and the architectural choices that make them tractable (precomputing hydraulics once, exploiting submodularity with a greedy search, using topology and elevation without a hydraulic model) all transfer to the SA Water network. The closest match is Ninh et al. (2025), developed in Adelaide for gravity sanitary sewers with small reticulation pipes, which turns blockage detection into a coverage condition and outperformed a utility's experience-based layout by about 43% in covered pipe length. Its coverage condition is the most direct starting point for the project's own objective, provided the typical depth it treats as fixed is allowed to rise with growth.

Second, what a sensor covers depends on what it is asked to detect and how. Level rises upstream of a blockage are reliable and falls downstream are not; detection weakens with distance, so that spacing beyond about 100 m degrades identification in small sewers; and steep terrain sharply limits what any level sensor can see. The project's evaluation measure should therefore be detection of simulated blockage and growth-driven surcharge events, not information, reconstruction error or covered length alone.

Third, sensors are neither uniform nor interchangeable. Flow sensors fail in small, multi-inlet maintenance holes; cheap ranging sensors can fail silently; level sensors can deliver flow through soft sensors; and binary switches carry useful information when their thresholds suit the chamber. The placement method should weight candidates by expected reliability and should treat the choice of sensor type as part of the decision rather than a given.

Fourth, the evidence that is most persuasive to a utility is operational. Sydney Water's experience provides a constraints checklist and a value case, I/I indicators computed from existing pumping-station records provide a no-cost first screen, and the five-point condition scale provides a language asset managers already use. The project's recommendations will be strongest if they are expressed in those terms. The two clearest gaps the project can address are placement under a changing load distribution and placement of a mixed set of sensor types, both evaluated against detection of the overflow events SA Water needs to prevent.

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